

Automated Ship Classification in Remote Sensing Imagery Using Knowledge Distillation Feature Maps

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ABSTRACT

Automated ship classification of maritime vessels operating in littoral zones are paramount for effectively safeguarding national and international maritime interests related to ship traffic. Over recent years, different deep learning approaches have made notable progress in image analysis and classification. Still, it requires significant processor power and memory, which poses deployment challenges for such a system on satellites and edge devices. To address this, a lightweight deep learning network is designed to minimise computational parameters and complexity while retaining a high recognition rate in the ship classification task. The proposed method introduces the MoBile Coordinated Attention (MBCA) block, which replaces the conventional Squeeze-and-Excitation with Coordinate Attention to preserve spatial features during knowledge distillation. This study focuses on the binary classification of ships using 4,000 satellite images to address the latency and bandwidth bottlenecks. Experimental results show that the proposed model achieves a classification accuracy of 97.25 %, outperforming standard architectures such as ResNet-50 and EfficientNet-B0 while reducing the parameter count to about 2.35M.

Keywords: Ship classification; Remote sensing imagery; Knowledge distillation; MobileNet; Coordinate attention

1. INTRODUCTION

Autonomous ship classification technology is advancing rapidly from testing to real-world applications. It has the potential to reduce challenges caused by human navigation errors, thereby lowering labour costs, improving navigation safety, and increasing profit margins. It assists in maintaining maritime security, monitoring illegal activities in oceans, and supporting effective decision-making in naval operations for coastal surveillance¹. Remote sensing satellites offer effective methods for tracking ships at sea². These satellites capture optical remote-sensing images and transmit them to ground stations for further analysis. Ground-based systems extract information from these images using artificial intelligence-enabled computers to simplify the task of image analyst³. However, bandwidth limitations between satellites and ground stations cause significant delays when processing large volumes of images⁴. Deep learning image processing systems can be installed on board devices to address this bottleneck.

Convolutional neural networks (CNNs) have emerged as a powerful approach for detecting and classifying ships because they can automatically learn discriminative features from large volumes of images or datasets⁵. However, deploying image processing systems directly on satellites presents challenges due to their substantial memory and processing requirements⁶. Traditional deep learning models are often computationally demanding, requiring significant memory and processing

power⁷. It makes them impractical for deployment on mobile, small satellites, IoT sensors, and autonomous drones. To address this challenge, researchers have focused on deploying lightweight deep learning models that balance accuracy on resource-constrained devices⁸. MobileNets are lightweight, efficient convolutional neural networks (CNNs) used on edge devices such as mobile and embedded systems. Their main focus is on depthwise separable convolutions (a combination of depthwise and pointwise convolutions). It reduces the number of network parameters, requiring less computational power than traditional CNNs.

Employing the knowledge distillation (KD) technique can further lower the computational complexity of the lightweight deep learning model⁹. It transfers knowledge from a pre-trained, high-capacity, complex model to a lightweight model. It considers the knowledge as the input-output mapping learned by a teacher (base) model and transfers it by training a lightweight student (target) model to mimic the teacher's outputs. Different distillation techniques have recently been applied to ship image classification tasks¹⁰⁻¹¹. Offline knowledge distillation utilises a pretrained and fixed-parameter base model to guide the target model. In contrast, the online knowledge distillation method trains the base and target models together from scratch by simultaneously updating their network parameters. Self-distillation uses a single network that serve as both the target and the base model. Therefore, choosing an appropriate architecture for both teacher (base) and student (target) is crucial to optimising the efficacy of the knowledge distillation approach.

Although existing lightweight and knowledge distillation-based methods effectively reduce computational complexity, previous work relies on channel-wise attention mechanisms such as Squeeze-and-Excitation (SE). It utilises global average pooling, which is beneficial for channel recalibration. It compresses two-dimensional spatial information into a one-dimensional representation, thereby losing important spatial information. It negatively hampers ship detection tasks, where capturing precise orientation and elongated vessel structures is essential. To overcome this limitation, the proposed methodology introduces the core approach of the MoBile Coordinated Attention (MBCA) block, which constitutes the fundamental building unit for both the base and target networks.

It is designed to facilitate a seamless feature distillation process, in which the target model learns from the teacher's intermediate features. It ensures the efficient knowledge transfer without information loss between two different architectures. The proposed method incorporates the Coordinate Attention (CoordAtt) module to enhance the representational power of the lightweight model. Classical attention mechanisms often degrade spatial information using global pooling¹²⁻¹³. In contrast, CoordAtt is designed to preserve precise positional information by embedding coordinate-aware attention. It achieves this by factorising global pooling into two separate 1D pooling operations along the vertical and horizontal axes, which allows the network to acquire long-range spatial dependencies and precise location cues simultaneously. The resulting attention maps re-weight the feature map based on channel importance and spatial coordinates. This efficiency and preservation of location make CoordAtt highly effective for tasks such as satellite image analysis on resource-constrained edge devices, where identifying the exact positions of objects, such as ships is critical¹⁴.

1.1 Contributions of the study

The main goal of the research study is to devise a lightweight yet effective deep learning model for ship classification in remote sensing images, while retaining spatial and positional information. The study further aims to improve the effectiveness of knowledge distillation between teacher and student models and to reduce computational complexity for deployment on resource-constrained devices. The major contributions of the research work are outlined as follows:

- A novel MoBile Coordinated Attention (MBCA) block is introduced, which incorporates coordinate attention into a lightweight MobileNet-based model to retain positional and spatial information for effective ship classification.
- A feature-level knowledge distillation framework is implemented to efficiently transfer spatially rich representations from a heavyweight teacher model to a compact, lightweight student model.
- The proposed methodology achieves an optimal balance between accuracy and computational parameters, making it suitable for deployment on resource-constrained satellite platforms.
- The proposed method outperforms existing lightweight and conventional deep learning models across classification accuracy, parameter efficiency, and inference time.

Although MobileNet, coordinate attention, and Knowledge Distillation have been studied independently in prior research, the proposed approach is not merely a straightforward combination of these techniques. Instead, it introduces a problem-driven MBCA block that replaces the conventional SE-based attention in the MobileNet architecture to address spatial information loss effectively. This design preserves directional spatial dependencies, which are essential for accurately classifying elongated orientation-sensitive objects such as ships, thereby distinguishing the proposed method from existing approaches.

The rest of this paper is structured as follows: Section II gives a brief discussion of existing research work; Section III explains the Proposed Methodology; Section IV gives Experimental result analysis, followed by the final concluding remarks and future directions in Section V.

2. LITERATURE REVIEW

Autonomous ship classification in remote sensing imagery using deep learning methods has emerged as a pivotal research area. The literature review will explore cutting-edge technologies for identifying maritime vessels under diverse environmental conditions and across sensor modalities. A deep CNN-based framework followed by a region proposal network (RPN) was proposed that identified ships using anchors shaped by ship geometry¹⁵. A hybrid deep learning network was developed to extract the meaningful information¹⁶. The extracted feature space was subsequently used to train a k-Nearest Neighbour (kNN) classifier to determine the presence of a ship. Similarly, the study presented the deep CNNs for effective ship detection and classification¹⁷.

Deploying deep CNN-based models for real-time ship detection on satellites is limited by high computational costs, making them unsuitable for resource-constrained platforms such as satellites and edge devices. MobileNet is a popular lightweight model designed to be small and efficient meeting the speed (latency) and size requirements of their application¹⁸. MobileNet is a lightweight module that effectively optimises feature extraction using convolutional block attention without sacrificing detection accuracy, enabling effective deployment on edge devices. The suggested study presented three variants of the MobileNet models (Thin, Balanced, and Shallow) for the classification task¹⁹. Thin MobileNet employed dropout, random erasing, layer pruning and separable convolutions to minimise overhead.

Balanced MobileNet retained additional layers with similar regularisation and convolutional improvements. Shallow MobileNet achieved faster inference by utilising the blockreduced design leveraging drop activation, random erasing, and separable convolutions. The proposed variants outperformed the baseline MobileNet model on the CIFAR-10 dataset. The proposed study evaluated the performance of the MobileNetv1 model on the ARM-based ms8pro micro device²⁰. The method obtained the 92.40 % classification accuracy using 2.1 W of power on the Caltech101 dataset. Recent studies show that channel attention mechanisms like Squeeze-and-Excitation increase the mobile network performance; they overlook spatial positional information.

Coordinate Attention has been introduced to overcome this by incorporating positional context information into the channel attention mechanism. Thus, coordinate attention is lightweight, easily integrable into mobile networks (e.g., MobileNetV2, EfficientNet), and improves performance for classification, object detection, and segmentation tasks. Despite these technical advantages, lightweight models often experience a degradation in classification accuracy compared to their heavyweight counterparts.

Knowledge Distillation (KD) presents an effective solution to bridge this performance gap by knowledge transfer from a heavyweight teacher model to a lightweight student model. A lightweight detector with a customised backbone and network pruning method was developed to minimise model capacity size and floating-point operations (FLOPs)²¹. Also, the knowledge distillation method was employed to transfer inter-level feature relationships from a larger, high-capacity base model to the compact target model. Similarly, the teacher-student model was implemented using an online and offline distillation scheme²². The proposed study implemented a novel SHAKE method that added a lightweight shadow head on the backbone architecture. The method combined the efficacy of the online and offline knowledge distillation approaches to demonstrate the superior performance of SOTA models. Similarly, the presented study proposed a multi-stage knowledge distillation method that combined the cross-stage feature map fusion, attention-based concatenated feature maps and contrastive loss to align the student and teacher intermediate feature map representation²³. The discussed method improved the classification accuracy from 69.06 % to 71.34 % and from 66.54 % to 68.03 % on the respective CIFAR100 and TinyImageNet datasets.

The proposed ARSD framework improved lightweight models using multiscale feature imitation and strict regression distillation, achieving superior performance over existing KD methods²⁷. The method obtained 68.28 % and 70.10 % mAP on DOTA and DIOR datasets. Similarly, the method outperformed the existing methods by showing the least 13.08M and 13.10M parameters on the respective datasets. The study introduced spatial-and channel-oriented structure discriminative modules (SCSDM) for effective feature extraction while suppressing noise²⁸. The suggested method outperformed the SOTA methods, achieving 92.00 % mAP on the RSOD dataset with a compact 3.3M parameters and high inference speed of 588.20 FPS. However, despite these technical advancements, the existing approaches primarily focus on feature selection, regression alignment, or noise suppression, while overlooking the preservation of spatial and positional information during feature learning.

3. MATERIALS AND METHODS

3.1 Preliminary Study

Knowledge distillation serves as a powerful method for model compression while maintaining comparable performance. In the teacher-student framework, a pre-trained larger model (teacher or base) generates soft output probabilities, which effectively guide the training of a smaller and more efficient model (student or target). Knowledge distillation offers enhanced architectural flexibility, allowing

for structural differences between the base and target models. Softmax with temperature equation is used to “soften” the outputs (logits) of both the base and target models, as indicated in Eqn. 1.

$$\sigma(p_i, T) = \frac{\left\{ e^{\frac{p_i}{T}} \right\}}{\left\{ \sum_j e^{\frac{p_j}{T}} \right\}} \quad (1)$$

Here, $\sigma(p_i, T)$ the softened probability for the i^{th} class. p_i is the logit (raw model output) for the i^{th} class. T is the temperature.

Distillation loss (soft loss) is the core of the knowledge transfer. It measures how different the student’s softened outputs are from the teacher’s softened outputs. It is typically the Kullback-Leibler (KL) divergence between the two distributions. The T^2 term is a scaling factor to ensure the gradient magnitudes from the soft and hard targets are roughly on the same scale, as depicted in Eqn. 2.

$$L_{soft}(p_t, p_s) = T^2 \cdot KL(\sigma(p_s, T) \parallel \sigma(p_t, T)) \quad (2)$$

Here, L_{soft} is the distillation loss. p_t and p_s are the logits from teacher and student model respectively. $KL(P \parallel Q)$ is the KL divergence, measuring the difference from distribution Q to P.

The student or target model is trained using a weighted sum of two losses: the distillation loss (learning from the teacher) and a standard supervised loss. This ensures the student model to learn the nuanced “dark knowledge” from the teacher while still being grounded by the correct answers, as indicated in Eqn. 3.

$$L_{total} = (1 - \alpha) \cdot L_{hard}(y, \sigma(p_s, T = 1)) + \alpha \cdot L_{soft}(p_t, p_s) \quad (3)$$

where, L_{total} is the final loss function to be minimised. L_{hard} is the standard supervised loss against the true labels. $\mathcal{Y}$ are the one-hot encoded true labels. L_{soft} is the distillation loss. α is a hyperparameter (between 0 and 1) that balances the two loss terms.

In the proposed knowledge distillation methodology, the ResNet-50 pre-trained model is used as the teacher network due to its strong feature representation capability. The teacher model consists of almost 24.97 million parameters. It is initialized with pretrained weights and further adapted on the ship dataset for binary classification. The lightweight student model is based on the proposed MobileNet architecture integrated with the MBCA block. The teacher model is subsequently used to guide the student model through soft target predictions and feature-level knowledge transfer.

3.2 Problem Formulation

Ship classification is the task of automatically identifying the presence of ships from satellite, aerial or remote imagery. Assume the remote sensing imagery ship dataset is characterised by $D=(G, L)$ where, $G=G_1, G_2, G_3, \dots, G_n$ and $L=L_1, L_2$ labels. It includes a set of n image instances categorised into two different class labels. The input image $I \in \mathbb{R}^{H \times W \times C}$ illustrates the ship images, where $H=W$ the spatial extent of the image is defined while C specifying the channel information. Given a ship image, the objective of this study is to develop a deep learning-based lightweight model M_θ for binary classification.

The proposed framework determines the class label ($\hat{L}_c$) for each image (G) by processing the feature maps F_c obtained from the input. The proposed work focuses on developing a lightweight student model for image-level ship classification, aiming to detect the presence of ships in images without using bounding-box localization.

3.3 Methodology

MobileNets are ideal for energy-constrained devices such as remote sensing, drones, and UAVs due to their low inference time and small model size. It represents a novel approach to ship classification, combined with attention-based student networks optimised via knowledge distillation. A key innovation in MobileNet is the replacement of the standard convolutions, which have a computational complexity of $O(D_k^2 \times M \times N)$ with the depth-wise separable convolutions having a smaller complexity of $O(D_k^2 \times M + M \times N)$, where, D_k , M , and N represent the kernel size, input channels, and output channels, respectively.

Attention mechanisms can incorporate Squeeze-and-Excitation (SE) blocks to enhance feature learning and adaptively recalibrate channel weights to focus on important features. The squeeze operation enables the aggregation of global feature information, as indicated in Eqn. 4.

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j) \quad (4)$$

Here output z_c is associated with the c^{th} channel. Excitation captures channel-wise dependencies in detail, as shown in Eqn. 5.

$$\hat{X} = X \cdot \sigma(T_2(\text{ReLU}(T_1(z)))) \quad (5)$$

Here, X and z indicate the input feature map, and the global average-pooled descriptor. T_1, T_2 represents the learnable linear transformations, and $\hat{X}$ signifies attention-enhanced output.

The lightweight student model retains the self-attention mechanism but significantly compresses its architecture by reducing the dimension of the Query-Key-Value (QKV) projections and the output transformation layers. It leads to fewer parameters and lower computational cost, while maintaining strong performance through effective knowledge distillation from a high-capacity teacher.

3.4 CoordAtt Block

CoordAtt (coordinate attention) block functions as a computational module intended to strengthen the representational capacity of features learned in mobile networks. It takes an intermediary feature input tensor $X = [x_1, x_2, \dots, x_c] \in \mathbb{R}^{H \times W \times C}$ and produces a tensor $Y = [y_1, y_2, \dots, y_c]$ as output with augmented representations of the same spatial dimensions as X . To encode directional information, pooling operations with $(1, W)$ and $(H, 1)$ kernel sizes have been applied to the input X along the respective horizontal and vertical directions for each channel. Therefore, the output corresponding to the c^{th} channel at spatial position h along the height dimension and position w along the width can be expressed as Eqn. 6 and Eqn. 7, respectively.

$$z_c^h(h) = \frac{1}{W} \sum_{0 \leq i \leq W} x_c(h, i) \quad (6)$$

$$z_c^w(w) = \frac{1}{H} \sum_{0 \leq j \leq H} x_c(j, w) \quad (7)$$

Then, features have been aggregated independently across the horizontal and vertical axes through

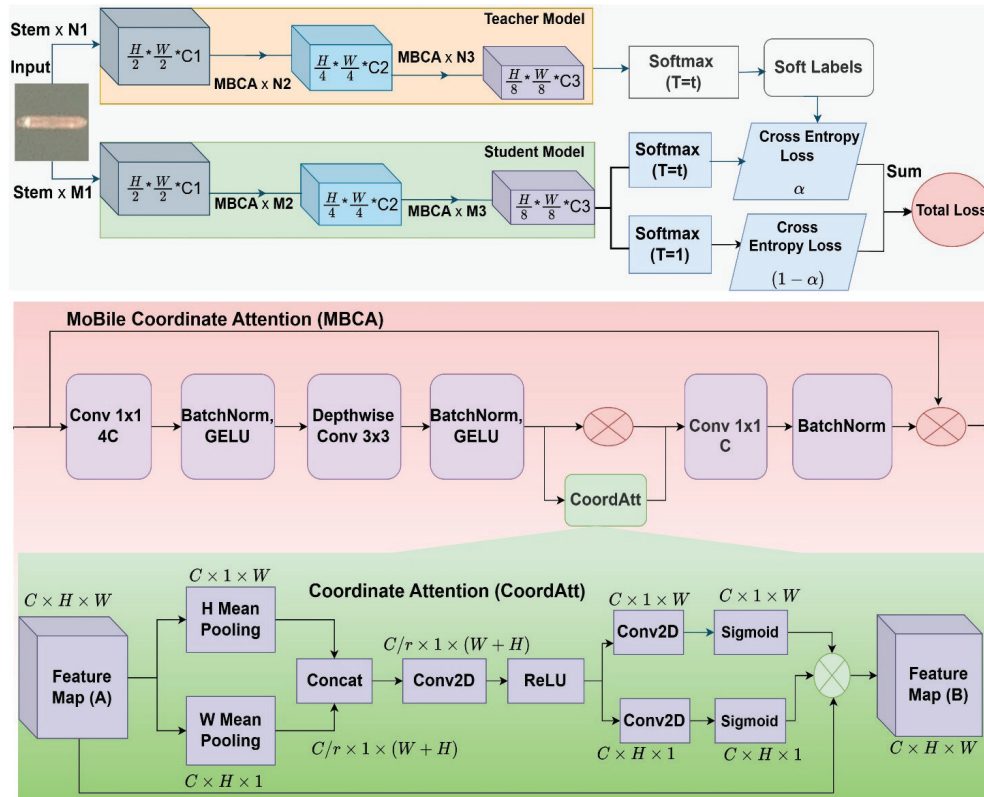


Figure 1. Overview of the proposed MBCA-based KD methodology. It illustrates the coordinate attention mechanisms for spatial and positional feature preservation.

these two transformations, resulting in direction-specific feature maps. The intermediate feature map can be denoted as $\mathbb{R}_r^{\mathbb{C}_{(H+W)}}$ and is obtained after encoding horizontally and vertically ($f^h \in \mathbb{R}_r^{\mathbb{C}_{\times H}}, f^w \in \mathbb{R}_r^{\mathbb{C}_{\times W}}$). The reduction ratio is denoted by r , which is used for controlling the block size.

Figure 1 shows the architecture of the proposed framework. The methodology makes use of the architecture of MoBILE Coordinate Attention (MBCA), which combines MBConv and coordinated attention and has fewer parameters. The MBCA (MoBILE Coordinated Attention) block is an improved, fully convolutional block based on the MobileNetV3 MBConv block. The main innovation is replacing the original SE module with a more advanced Coordinate Attention (CoordAtt) module. While the original SE module provides effective channel attention—helping the model determine “what” features are important—it does so by collapsing all spatial data, losing precise positional information about “where” those features are located. In contrast, the CoordAtt module is designed to address this by incorporating positional information into the attention mechanism. It captures long-range dependencies and precise location by encoding features across row-wise and column-wise dimensions separately. By integrating this position-sensitive attention, the MBCA block offers more powerful and comprehensive feature refinement than the original MBConv, making it especially suitable for tasks where spatial location is crucial.

4. EXPERIMENTAL RESULTS AND DISCUSSION

The following subsection gives insights into the ship dataset used for experimental result analysis. Additionally, the performance analysis of the proposed methodology has been further discussed to demonstrate the contribution of CoordAtt with MobileNet. Experimental results validate the robustness of the proposed methodology by comparing the classification accuracy with the existing SOTA models.

4.1 Dataset Description

The Ship in Satellite Imagery dataset, publicly available on Kaggle, consists of 4,000 images with dimensions $80 \times 80 \times 3$ (RGB colour channels). These images were captured by the PlanetScope satellite and feature a 3-meter spatial resolution per pixel²². The dataset is divided into two classes: 3,000 “No-Ship” images and 1,000 “Ship” images. The “Ship” class contains fully visible vessels of varying sizes, orientations, and environmental conditions, ensuring diversity in the training and testing samples. The dataset has been divided into train, validation, and test samples with the ratio of 6:2:2, as shown

Table 1. Ship imagery dataset description showing the number of ship and no-ship instances across training, validation, and testing sets.

Classes	Training	Validation	Testing
Ship	600	200	200
No Ship	1800	600	600
Total	2400	800	800



Figure 2. Sample images from the dataset illustrating the two classes: ship and no-ship.

in Table 1. Therefore, the training, validation, and testing datasets contain a total of 600, 200, and 200 ship instances, respectively, and 1800, 600, and 600 non-ship instances, respectively. Figure 2 represents the instances of the images containing Ship and No-Ship.

4.2 Performance Comparison And Result Analysis

The proposed lightweight model shows the lowest parameters and low processing cost while maintaining acceptable accuracy. Notably, it outperforms the other SOTA models and is more efficient in terms of computational complexity as measured as FLOPs or Mega Multiply–accumulate operations (MMac). Table 2 shows the experimental outcomes of the different SOTA and proposed methods. The proposed methodology has obtained the highest classification accuracy of 97.25 %, while the pre-trained ResNet-50 has achieved the least accuracy of 92.71 %. Therefore, the proposed methodology has obtained a marginal improvement of approximately 4.54 % over the least performing ResNet model. Similarly, the proposed model is the lightweight model by achieving the fewest parameters (2.35M). On the flip side, VGG-16 is the heavyweight model having the highest parameters (133.23M). The suggested MBCA method has shown the least inference time (23.50 ms), indicating that it is well-suited for real-time applications.

Moreover, an ablation study has been carried out to assess the contribution of each architectural component in the proposed methodology, as shown in Table 3. The baseline MobileNet model achieves the least classification accuracy of 94.12 %, which significantly improves to 95.68 % with the addition of knowledge distillation, showing the effectiveness of teacher-student learning. Integrating SE attention provides a marginal improvement (95.01 %), indicating limited capability in capturing spatial information. In contrast, the proposed MBCA module further increases the performance to 96.34 % by retaining positional and spatial information. Finally, the optimal balance between accuracy and computational efficiency has been achieved by integrating MBCA and KD, yielding the highest classification accuracy of 97.25 % with only a slight increase in parameters.

4.3 Result Visualisation

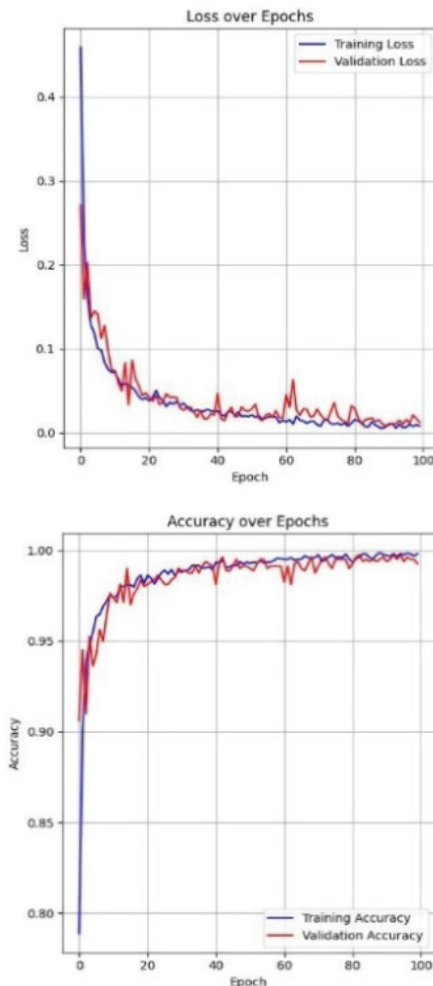
Figure 3 reflects the accuracy and loss curves for training and validation datasets of ship and no-ship images till 100th epochs. The steadily increasing accuracy curve indicates effective feature learning, while the decreasing loss curve

Table 2. Comparative result analysis of the proposed approach with existing SOTA models

Models	F1-Score	Accuracy (%)	MMac	Inference time (ms)	Parameters (in Millions)
VGG-16 ²⁴	0.8792	92.83	1923.2	172.6	133.23
ResNet-50 ²⁵	0.9123	92.71	1620.3	102.1	24.97
EfficientNet-B0 ²⁵	0.9354	95.72	709.9	83.0	5.68
MobileNet-SE ¹²	0.9402	96.49	472.7	39.7	4.96
MBCA (Proposed)	0.9439	97.25	229.09	23.5	2.35

Table 3. Ablation study showing the impact of knowledge distillation and attention mechanisms used in the proposed method

Models	KD	Attention	Accuracy (%)	F1-Score	Parameters (in Millions)
Baseline MobileNet	x	x	94.12	0.9120	2.10
MobileNet + KD	√	x	95.68	0.9276	2.10
MobileNet + SE	x	√	95.01	0.9219	2.18
MBCA (Proposed)	√	√	97.25	0.9439	2.35

**Figure 3. Training and validation loss and accuracy curves showing stable convergence with decreasing loss and increasing accuracy.**

demonstrates a reduction in error between predicted and actual labels as the epoch progresses. Also, the model converges 100th by visualising the slight fluctuations or a flat plateau in the loss and accuracy graph. The marginal difference between

validation and training dataset curves indicates the strong generalisation and minimum over-fitting of the lightweight student model.

The performance of the proposed approach has been further evaluated using RoC (receiver operating characteristic) curves, as shown in Fig. 4. It graphically represents the correlation between the True Positive Rate (TPR) and False Positive Rate (FPR) across varied decision thresholds. It shows how well the lightweight target model approximates the high-capacity base model's reaction. Despite having fewer parameters, a well-distilled lightweight target model can still achieve strong class separability, as reflected in a high area under the RoC curve.

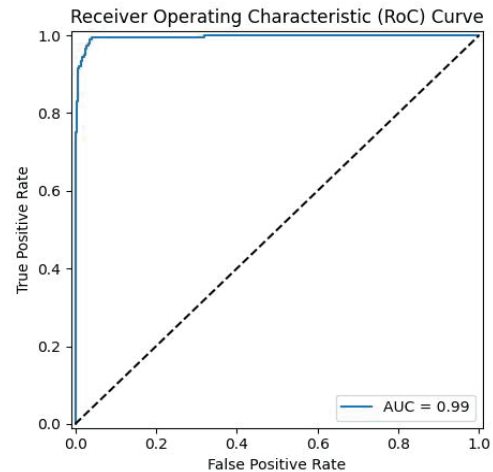
**Figure 4. RoC curve visualisation illustrating the proposed method's high classification performance with an AUC close to 1.0.**

Figure 5 shows the PR (precision-recall) curve, which illustrates the relationship between sensitivity and precision at varying threshold levels. The area under the PR curve provides a single numerical metric that summarises the collective effectiveness of the models. The figure shows a higher area under the curve, suggesting a better precision-recall trade-off,

making it particularly useful for tasks like object classification, where false positives or false negatives carry different costs.

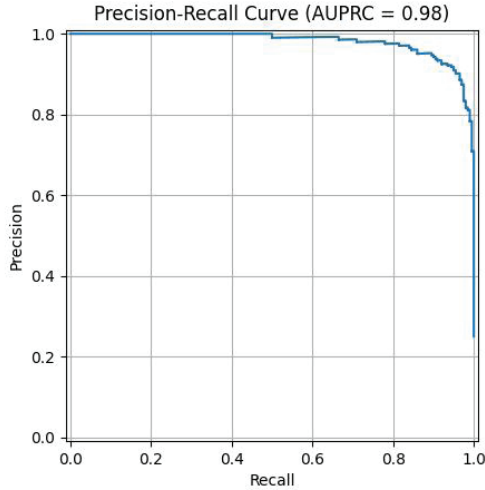


Figure 5. PR performance curve analysis showing strong model performance with high precision across varying recall levels (AUPRC = 0.98).

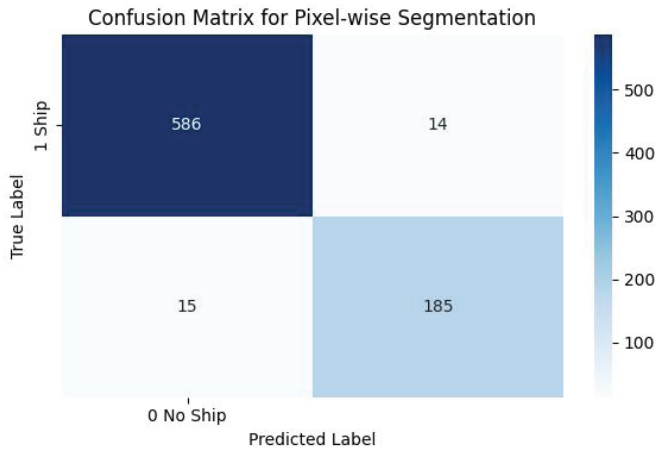


Figure 6. Confusion matrix visualisation showing high classification accuracy with minimal misclassification between ship and no-ship classes.

Figure 6 illustrates the qualitative analysis of the misclassified instances from the confusion matrix. The confusion matrix compares the actual images with the predicted images with the highest classification. However, some of the total instances have been misclassified as the “no-ship” category. Similarly, some instances have been misclassified as the “ship” category. It infers from the confusion matrix that errors mainly occur in two scenarios: (i) small vessels where the pixel count is insufficient for the student model to identify distinct features, and (ii) ships situated in high-clutter zones near port infrastructure. Although the teacher model offers guidance in these areas, the significant parameter reduction within the student model leads to occasional sensitivity to background noise.

4.4 Discussion

The proposed methodology is well-suited for defence Intelligence, Surveillance, and Reconnaissance (ISR)

applications, where strict constraints on latency, power, and bandwidth are critical. In such a scenario, on-board processing systems must operate with limited computational power and constrained communication bandwidth between satellite and ground stations. Table 2 illustrates that the developed lightweight MBCA-based framework has achieved a low computational cost of 229.09 MMac, an inference time of 23.50 ms, and a lightweight network with 2.35M parameters, enabling efficient on-board processing and reducing the need for high-bandwidth data transmission to ground stations.

The proposed method ensures reliable and robust classification of ships in complex maritime environments, making it suitable for real-time ISR operations in Defence applications. Furthermore, the proposed method achieves strong performance on the benchmark dataset; its robustness under diverse challenging conditions has not been explicitly stated. It may hamper model performance in real-world maritime scenarios, though integrating of coordinate attention into the MBCA block can help preserve spatial features and improve robustness. Additionally, cross-dataset validation has not been performed in the current study, thus limiting the assessment of its generalization capability across diverse remote sensing datasets. This will be considered as an important direction for future work.

Although the present study focuses on binary ship vs no-ship classification to establish the effectiveness of the proposed MBCA-based lightweight framework, this formulation serves as a fundamental step towards more complex maritime analysis tasks. Binary classification is particularly relevant for real-time surveillance scenarios where rapid identification of ship presence is critical before performing detailed categorization, such as vessels, tankers, and naval. Compared to existing methods, which primarily focus on feature alignment, noise suppression, and multiscale learning, the proposed approach emphasises the preservation of spatial and positional information through coordinate attention. This provides a significant advantage in ship classification tasks where object orientation and structure are critical. Furthermore, unlike SE-based lightweight models, which lose spatial information, the proposed MBCA block maintains directional dependencies, resulting in improved performance at lower computational cost. Therefore, the MBCA architecture-based student model offers a good trade-off between complexity and speed, maintaining a relatively low number of parameters while substantially reducing FLOPS.

5. CONCLUSION

The proposed study addresses the fundamental challenge of designing computationally efficient yet highly accurate deep learning models for deployment in resource-limited devices, particularly on-board satellite systems. The primary contribution is the introduction of the MoBile Coordinated Attention (MBCA) block. This innovative lightweight convolutional module synergistically combines the computational efficiency of MobileNet architecture with the spatially aware Coordinate Attention (CoordAtt) mechanism. A novel knowledge distillation framework employing the MBCA block has been developed as a shared architectural component

across both teacher and student models. Experimental results confirm that our approach yields a lightweight student model having 2.35M parameters and 97.25 % classification accuracy.

However, the robustness of the proposed study under challenging real-world maritime conditions such as sea clutter, haze, varying resolutions, and partial occlusions has not been extensively evaluated. This limitation may affect generalization in practical deployment scenarios and will be the foundation for future work. The current study provides a binary ship classification baseline that can be further extended to multi-class and detection tasks in complex maritime scenarios. Future research will focus on enhancing tiny target detection in challenging weather conditions, including low-light night-time settings and unstable environments affected by waves and other external disturbances, and on exploring cross-dataset validation to assess generalisation performance across diverse remote sensing environments.

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